MPU6886 Gyroscope
Thermal Drift & Calibration Analysis

Internal R&D Report

Sensor	MPU6886 (6-axis MEMS IMU, no on-chip temp compensation)
Platform	M5Stack AtomS3 (ESP32-S3 + 0.85" IPS display)
Test sessions	7 runs — 3 series (display ON face-up / face-down / OFF)
Temperature range	28.1°C – 65.0°C
Total samples	~40,325 (11.2 h of data, 1 Sa/s)
Date range	2026-03-03 → 2026-03-06
Heating method	Hot water bag (~80°C) in sealed enclosure, natural convection
IR shielding	Paper ball insulation to block directional IR radiation
Analysis	DC-offset, Poly-3, LUT (0.5°C bins), dG/dT, integration simulation

This report characterizes temperature-dependent bias drift in the MPU6886 MEMS gyroscope mounted on the M5Stack AtomS3 platform. The primary use case is mobile telemetry on ground vehicles where the AtomS3/MPU6886 provides orientation and angular rate data. The report evaluates whether a pre-recorded thermal calibration (LUT or polynomial) can reliably compensate gyro drift at runtime, and identifies the conditions under which this approach fails.

Report generated 2026-04-02 15:10 UTC · Analysis: Python 3 / pandas / scipy / matplotlib

Table of Contents

1. Experimental Setup & Methodology

2. Raw Data & The Run-to-Run Bias Problem

3. LUT Construction & Honest Residual Assessment

4. Polynomial (Poly-3) Compensation Assessment

5. Per-Axis Verdict: GX, GY, GZ

6. Thermal Sensitivity dG/dT & Repeatability

7. Display / Screen Condition Effect

8. Integration Error Simulation

9. Conclusions — What Works and What Does Not

10. Recommended Approach for Telemetry Systems

Appendix A — Metric Definitions

Appendix B — Full Numeric Results

1. Experimental Setup & Methodology

The MPU6886 does not implement on-chip temperature compensation. Its internal die temperature sensor (register 0x41) provides a reliable proxy for die temperature but not for ambient. All 7 test runs are monotonic heating sequences — the device was placed inside a sealed enclosure and heated externally via a hot water bag (~80°C) providing energy through natural convection. Direct infrared radiation from the heat source was shielded using crumpled paper balls creating inhomogeneous air gaps between the source and sensor, preventing directional thermal gradients from IR irradiation. Conductive heat transfer was similarly blocked by this insulation layer. The AtomS3 processor and display (when active) contribute additional uncontrolled self-heating.

No forced cooling was applied, and no single session contains both a heating and a cooling phase. This is an important structural limitation: classical hysteresis (different response on heating vs cooling within the same session) cannot be directly measured. What can be measured is the run-to-run non-repeatability of the heating curve.

Pre-processing: all signals were smoothed with a Savitzky-Golay filter (window=21 samples, polynomial order=3) before any calibration analysis, to separate the thermal drift signature from sample-level noise. All residual metrics reported here are therefore conservative lower bounds — the raw (unsmoothed) signal will show higher errors.

Run	Condition	Date	T start	T end	N samples
T1-Su	ON (face up)	2026-03-03	35.4°C	57.8°C	5,732
T1-Giu	ON (face down)	2026-03-04	35.6°C	61.5°C	5,227
T1-Spento	OFF	2026-03-05	33.1°C	57.9°C	5,560
T2-Su	ON (face up)	2026-03-04	28.1°C	59.8°C	7,016
T2-Spento	OFF	2026-03-05	30.8°C	64.4°C	5,677
T3-Su	ON (face up)	2026-03-04	30.2°C	59.4°C	5,383
T3-Spento	OFF	2026-03-05	29.1°C	57.0°C	5,730

Run naming convention: *Run IDs use Italian abbreviations from the original test log. "Su" = face up (su = up), "Giu" = face down (giù = down), "Spento" = display off (spento = turned off). T1/T2/T3 denote the test series number.*

2. Raw Data & The Run-to-Run Bias Problem

The raw gyroscope output plotted against die temperature reveals two dominant error sources:

Absolute DC bias variability: each run starts at a completely different bias level. Differences of up to 2–3°/s between sessions are observed on GX. This is caused by stress relaxation, power-cycle settling, and thermal history at startup — all non-deterministic.

Monotonic thermal drift: within each run the bias changes continuously as the die heats. The total drift from 30 to 60°C exceeds 2°/s on GX and GY.

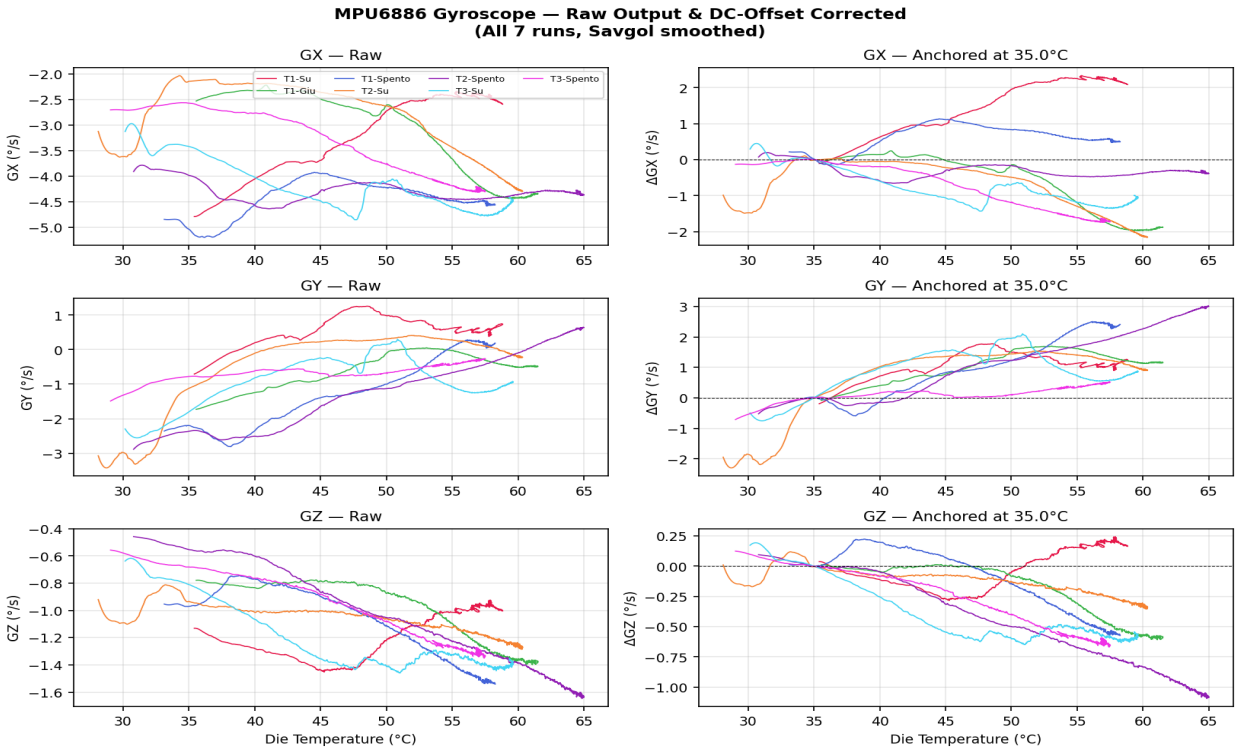

Figure 1 — Left: raw output all 7 runs (run-to-run DC bias clearly visible). Right: same data after anchoring each run at 35°C — thermal drift shape becomes visible, but curves still diverge significantly.

Axis	Mean σ (°/s)	Max σ (°/s)	Total drift (35→57°C)
GX	0.769	1.394	≈ 0.6 °/s
GY	0.459	0.660	≈ 1.4 °/s
GZ	0.174	0.293	≈ 0.4 °/s

The DC bias is the dominant error and must be measured at every power-on via a static calibration (ZARU — Zero Angular Rate Update). Without it, no compensation strategy is meaningful.

3. LUT Construction & Honest Residual Assessment

The temperature range is divided into 0.5°C bins; the consensus LUT is the element-wise median across all run-specific LUTs. This approach makes no assumptions about drift shape and is computationally trivial. Its validity depends entirely on the inter-run repeatability of the drift shape.

The residual is computed as: **residual = (raw – DC_offset_at_startup) – consensus_LUT(T)**, where DC_offset_at_startup is the mean raw value measured during the initial static calibration window, and consensus_LUT(T) is the median LUT from all training sessions anchored to zero at 35°C. This is exactly what firmware running on the device would compute.

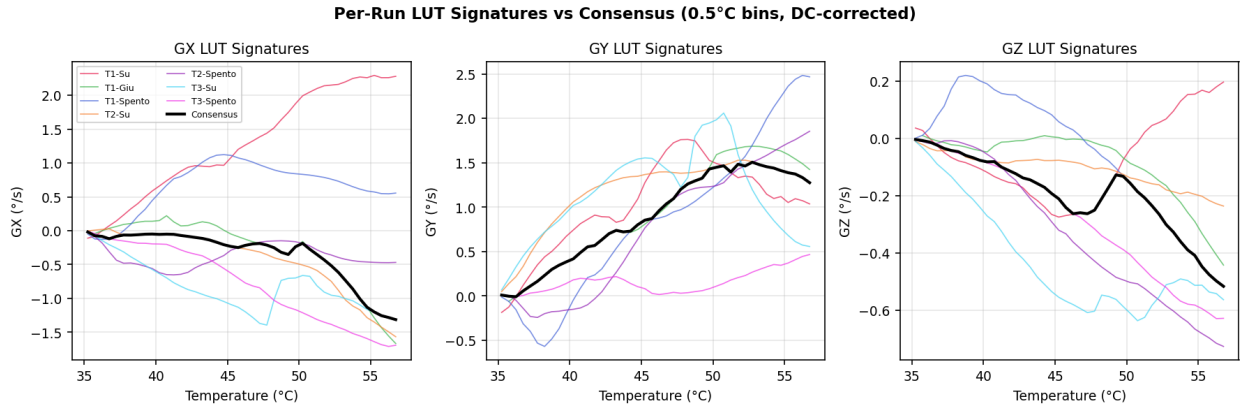

Figure 2 — Per-run LUT signatures for all three axes. Black line is the consensus. GX shows major divergence between runs. GZ curves are noticeably tighter.

Stability Threshold Reference

Threshold	Value (m°/s)	Effect on telemetry
Stable	< 50	Filter unaffected — transparent to any PID or EKF
Degraded	50 – 150	Slow yaw drift; heading error ≈ 5°/min
Unstable	150 – 500	Attitude estimation visibly affected over seconds
Unusable	> 500	Integration diverges; navigation loop unreliable

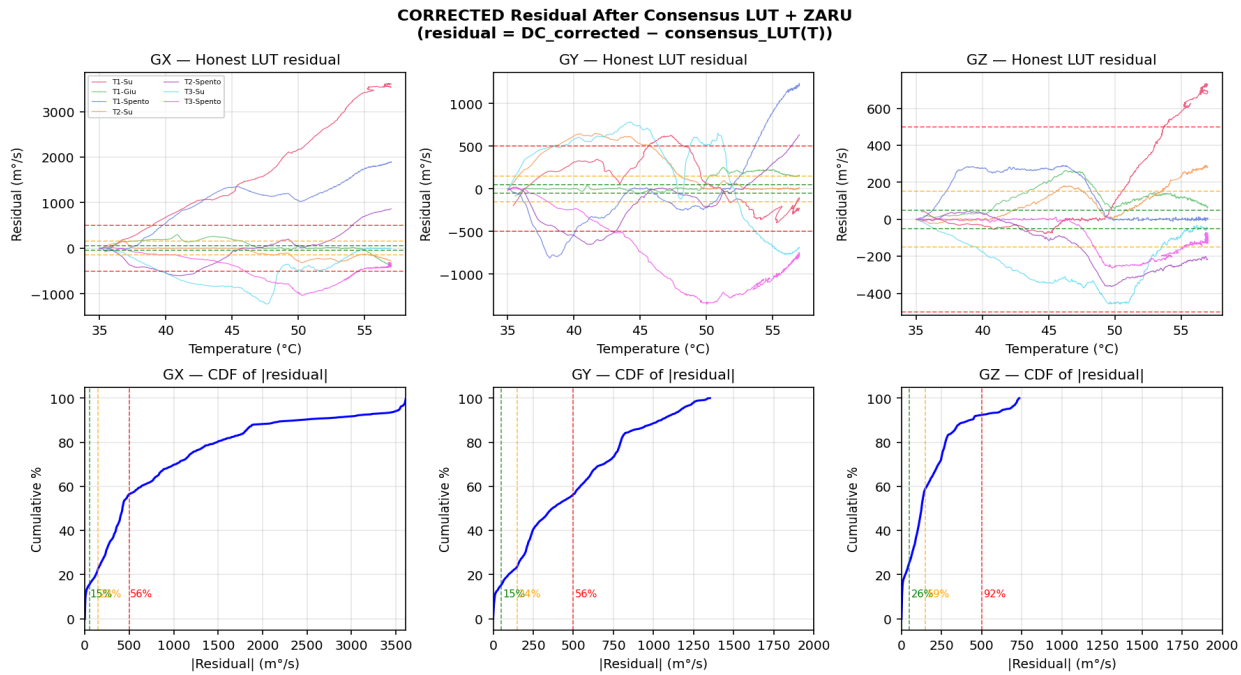

Figure 3 — CORRECTED residual after consensus LUT + ZARU. Top: residual vs temperature per run. Bottom: CDF of |residual| in 35–57°C range.

4. Polynomial (Poly-3) Compensation Assessment

A degree-3 polynomial is fitted to each run individually (on the DC-corrected signal vs temperature). The consensus polynomial is the coefficient-wise mean across all 7 per-run fits. The honest residual is computed identically to the LUT case: **residual = DC_corrected** – **consensus_poly(T)**. The device at runtime has access only to the consensus model, not to its own run-specific fit.

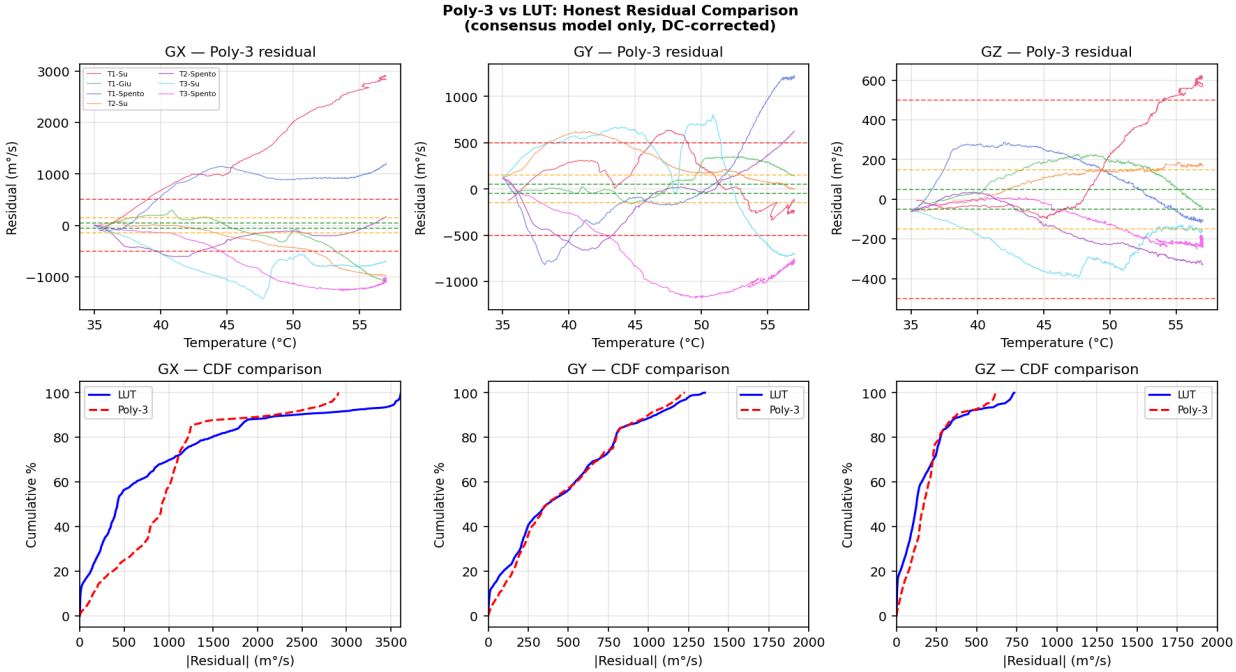

Figure 4 — Top: Poly-3 honest residual vs temperature. Bottom: CDF comparison of LUT vs Poly-3. Poly-3 is comparable or worse than LUT, confirming that the bottleneck is inter-run shape variability, not model type.

Per-sample residual statistics after consensus Poly-3 + ZARU (35–57°C range):

Axis	Mean |res|	Std	P95	% stable	% unusable
GX	975 m°/s	703 m°/s	2773 m°/s	3%	75%
GY	471 m°/s	344 m°/s	1134 m°/s	7%	43%
GZ	194 m°/s	145 m°/s	559 m°/s	15%	7%

The Poly-3 residuals are comparable to — or worse than — the LUT residuals on all axes. On GX, Poly-3 performs noticeably worse (mean |res| 975 m°/s vs 861 m°/s for LUT, 75% unusable vs 44%). This is because the polynomial, being a rigid parametric model, cannot track the consensus drift shape as closely as the non-parametric LUT — and yet even the LUT already fails. The conclusion is reinforced: the bottleneck is inter-run shape variability, not model flexibility. No fixed compensation model — whether parametric or non-parametric — can address a non-deterministic drift shape.

5. Per-Axis Verdict

Per-sample LUT residual statistics (35–57°C range). "Mean |res|" = mean absolute residual across all individual samples from all 7 runs. "Std" = standard deviation of per-sample residual.

Axis	Mean |res|	Std	P95	P99	Peak	% stable	% degraded	% unstable	% unusable
GX	861	999	3541	3607	3618	15%	7%	34%	44%
GY	462	371	1174	1291	1351	15%	8%	33%	44%
GZ	178	180	651	723	734	26%	33%	33%	8%

5.1 GX and GY: LUT Compensation Fails

For GX, 44% of samples fall in the "unusable" category after LUT correction (mean residual 861 m°/s). For GY, 44% are unusable (mean 462 m°/s). The LUT removes only the consensus shape, while the run-specific drift diverges by up to ±3.6°/s from that consensus on GX. Applying the LUT to a new run where the actual drift follows a different curve would actively **add error** rather than remove it in the worst case.

The root cause is the high dG/dT inter-run variability on GX and GY: the thermal sensitivity itself (°/s per °C of heating) is not repeatable between sessions. Different display states, different ambient start temperatures, and different self-heating rates all shift the drift curve shape. A single pre-recorded LUT cannot capture a non-deterministic shape.

5.2 GZ: Partially Usable

GZ is better: only 8% of samples exceed 500 m°/s (vs 44% for GX). Mean residual is 178 m°/s. 26% of samples are in the "stable" range. For non-critical yaw-rate estimation (turn detection, coarse heading) the GZ LUT provides a modest improvement. For precision heading integration it is still insufficient without an external correction source.

6. Thermal Sensitivity dG/dT & Repeatability

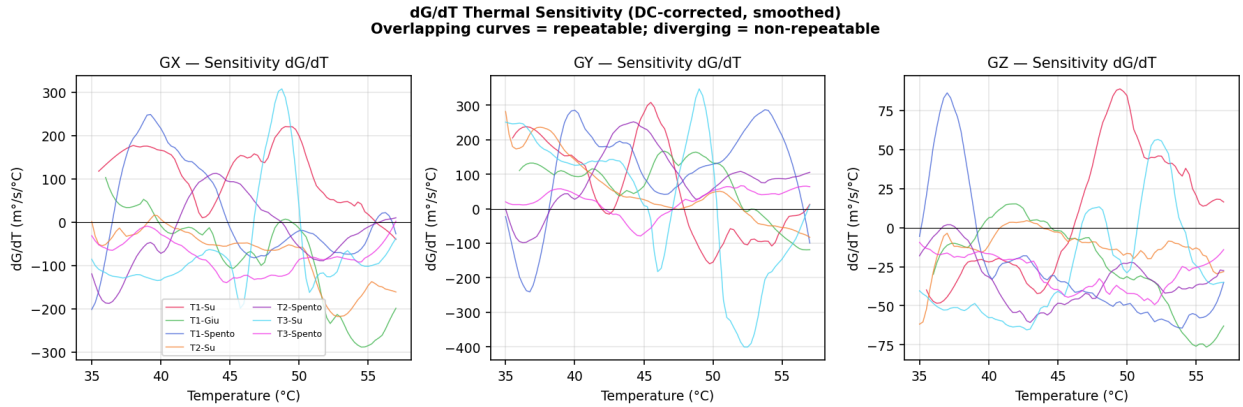

Figure 5 — dG/dT (thermal sensitivity) per run per axis. Overlapping curves indicate repeatable drift shape; diverging curves mean the shape itself varies between sessions.

Axis	Mean dG/dT (°/s/°C)	Inter-run σ of dG/dT	Max |dG/dT|	Verdict
GX	-0.0219	0.0987 °/s/°C	0.307 °/s/°C	Non-repeatable
GY	0.0642	0.1089 °/s/°C	0.401 °/s/°C	Non-repeatable
GZ	-0.0195	0.0294 °/s/°C	0.089 °/s/°C	Partially repeatable

An inter-run σ on the order of 0.1°/s/°C on GX means: if the sensor heats at 1°C/s, the instantaneous compensation error can be ±100 m°/s in a single second. Over a 25°C ramp this integrates to a total shape-mismatch of several °/s at peak divergence — matching the observed residuals.

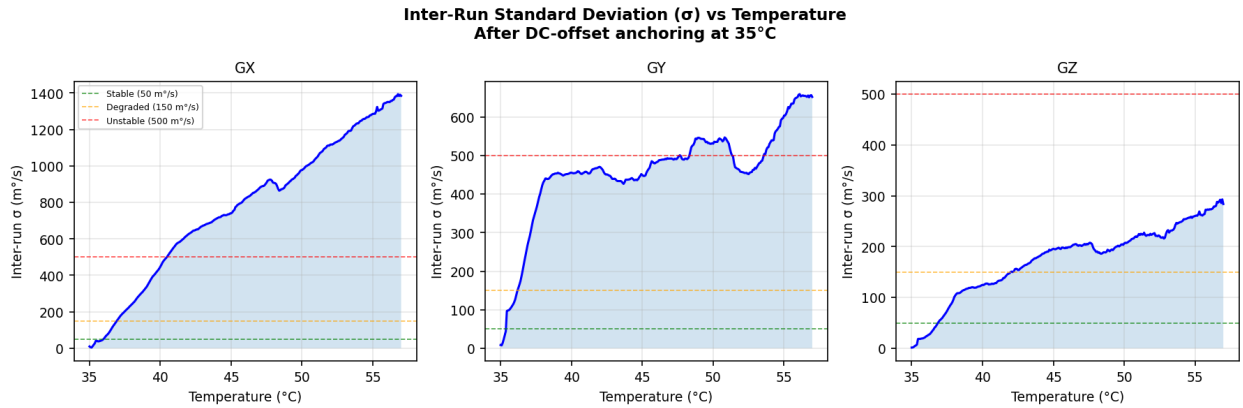

Figure 6 — Inter-run standard deviation (σ) vs temperature for each axis after DC-offset anchoring. The rising σ at higher temperatures reflects increasing divergence of drift shapes.

7. Display / Screen Condition Effect

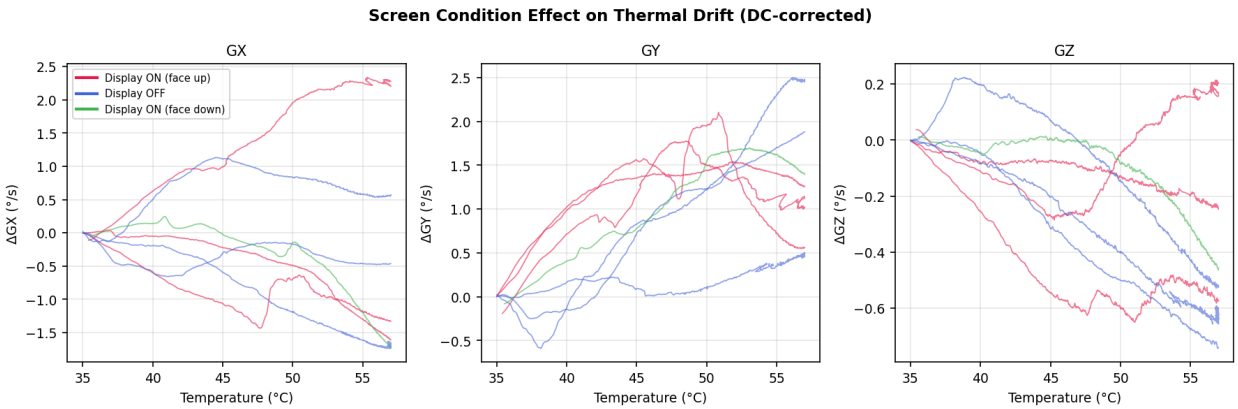

Figure 7 — DC-corrected output grouped by display state. Display-on and display-off sessions produce systematically different drift signatures, especially on GY.

The AtomS3 display is a significant thermal confound. When active, the backlight and LCD driver create a local heat source on one face of the PCB, introducing an internal thermal gradient that shifts the effective MEMS junction temperature relative to the internal register reading. This produces a systematic, session-class-specific offset in the drift shape — and is one of the main reasons GX and GY are non-repeatable across sessions.

The T1-Giu run (display on, face down) shows particularly anomalous behaviour: placing the display face-down reverses the convective gradient, creating a third distinct thermal signature. This is consistent with the hypothesis that drift variability is driven by physical heat flow geometry, not by sensor-intrinsic randomness — though further testing would be needed to conclusively confirm this.

Practical implication: even if a calibration LUT is built exclusively from display-on sessions (or exclusively display-off), the remaining session-to-session variability in ambient temperature and heating rate is still large enough to invalidate the LUT for GX and GY. The display effect worsens an already non-repeatable situation.

8. Integration Error Simulation

The simulation below models a 60-second warm-up from 30°C to 55°C — a representative scenario for a ground vehicle powering on cold. The spread between colored lines represents the actual range of outcomes across the 7 sessions.

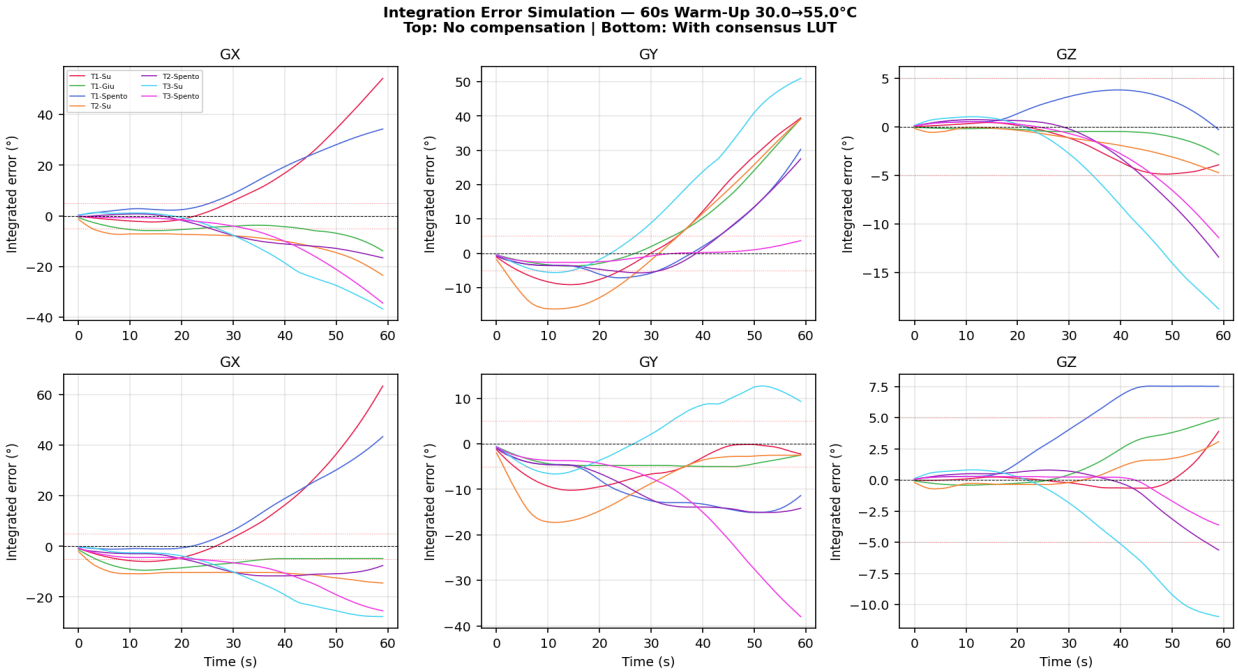

Figure 8 — Integration error simulation. Top: no compensation — errors can reach tens of degrees. Bottom: with consensus LUT — the spread between runs shows the residual uncertainty. GX/GY remain highly variable; GZ is noticeably better.

Key observation: with the LUT applied to GX and GY, individual runs may actually show larger integrated error than without the LUT — because the LUT subtracts the consensus shape but the actual run follows a different shape. This is the failure mode that cannot be predicted without knowing the run-specific drift in advance.

9. Conclusions — What Works and What Does Not

The MPU6886 gyroscope X and Y axes exhibit non-repeatable thermal drift that cannot be reliably compensated by a pre-recorded LUT or polynomial model. Applying such a model in production will not reliably reduce error and may increase it in sessions where the actual drift diverges from the trained consensus.

What Works

• **ZARU at power-on** — Mandatory. Removes the dominant error source (DC run-to-run bias). 10–20 seconds of static measurement. Zero additional hardware.

• **GZ axis with LUT** — Limited improvement. GZ shows ~4× lower shape variability than GX/GY. Useful for coarse yaw-rate estimation, not for precision heading integration.

• **Kalman filter with inflated Q** — Rather than fighting drift with a fixed model, allow the filter to track it by setting gyro bias process noise Q high. This lets the filter converge using GPS, wheel odometry, or magnetometer.

• **Display state consistency** — If a fixed display state is enforced, the systematic display-induced component is removed. Does not solve GX/GY, but reduces variability.

What Does Not Work

• **Pre-recorded LUT for GX and GY** — Drift shape not sufficiently repeatable. Inter-run range at some temperatures exceeds ±2°/s. This is a physics problem, not a data quantity problem.

• **Polynomial compensation for GX and GY** — Same limitation. The run-specific polynomial differs from the mean by the same amount as LUT divergence.

• **Any compensation without re-validation at each power cycle** — Drift shape can change between boots. Any pre-recorded model must be accompanied by a live DC correction at startup.

10. Recommended Approach for Telemetry Systems

The following architecture is the minimum viable approach for any telemetry system using the MPU6886 that requires reliable angular rate data over a self-heating temperature range.

[MANDATORY] ZARU at every power-on: hold static for ≥15 s, average gyro output, subtract. Record die temperature at calibration time.

[MANDATORY] Inflate the gyro bias process noise Q in your Kalman / complementary filter. Treat the bias as a slow-varying random walk.

[RECOMMENDED] Use an external yaw reference (GPS course, wheel odometry, magnetometer) as a correction input. Even 1 Hz is enough.

[RECOMMENDED] Fix the display state for all operational sessions. If headless, build calibration from display-off sessions only.

[OPTIONAL] (GZ only) A LUT or poly-3 model for GZ can provide marginal improvement in yaw-rate estimation. Apply on top of ZARU, never as replacement.

[AVOID] Do not apply a pre-recorded LUT or polynomial model to GX or GY.

[AVOID] Do not extrapolate the LUT outside the 35–55°C range.

[AVOID] Do not mix display-on and display-off sessions in the same training set.

For applications requiring heading accuracy better than ±5° over 60 seconds, the MPU6886 without an external reference is insufficient regardless of thermal compensation strategy. Consider a higher-grade IMU (e.g. ICM-42688-P or BMI088) with factory-calibrated temperature compensation for precision navigation tasks.

Appendix A — Metric Definitions

Two families of metrics are used in this report. They measure different things and should not be compared directly:

Per-sample residual (used in Section 5 tables): for every individual gyro sample in the 35–57°C range across all 7 runs, the residual is computed as *r = DC_corrected(t) – consensus_model(T(t))*. Statistics (mean |r|, std, percentiles) are computed over the pooled set of all ~30k samples. This represents what a real device would experience moment-by-moment.

Inter-run shape σ (used in Appendix B summary table): all 7 DC-corrected drift curves are resampled onto a common temperature grid. At each temperature bin, the standard deviation across the 7 runs is computed. "Shape σ mean" is the average of this σ(T) curve; "Shape σ max" is its peak value. This represents the irreducible uncertainty in the drift shape that any pre-recorded model must face.

The per-sample mean |residual| is typically higher than the shape σ mean because it also includes intra-run noise and the effect of non-uniform sample density across temperature. For GX, the difference (861 vs 769 m°/s) reflects the additional contribution of high-frequency noise within each run.

Appendix B — Full Numeric Results

All values use the corrected (non-circular) residual formula.

Table B1 — Inter-run shape variability *(σ of DC-corrected drift curves resampled on common temperature grid, 35–57°C):*

Method	Axis	Shape σ mean (m°/s)	Shape σ max (m°/s)
DC-offset only	GX	769	1394
DC-offset only	GY	459	660
DC-offset only	GZ	174	293

Table B2 — Per-sample residual comparison: LUT vs Poly-3 *(pooled across all runs, 35–57°C):*

Method	Axis	Mean |res| (m°/s)	Std (m°/s)	P95 (m°/s)	% stable	% unusable	Verdict
LUT (0.5°C)	GX	861	999	3541	15%	44%	fails
LUT (0.5°C)	GY	462	371	1174	15%	44%	fails
LUT (0.5°C)	GZ	178	180	651	26%	8%	marginal
Poly-3	GX	975	703	2773	3%	75%	fails
Poly-3	GY	471	344	1134	7%	43%	fails
Poly-3	GZ	194	145	559	15%	7%	marginal

Table B3 — Strategy summary:

Strategy	Axis	Status	Notes
ZARU only	GX	necessary	Removes DC bias; drift remains
ZARU only	GY	necessary	Removes DC bias; drift remains
ZARU only	GZ	necessary	Removes DC bias; drift remains
ZARU + LUT (GZ)	GZ	marginal	Mean |res| 178 m°/s
ZARU + KF(Q high)	All	recommended	Tracks drift in-flight via external ref